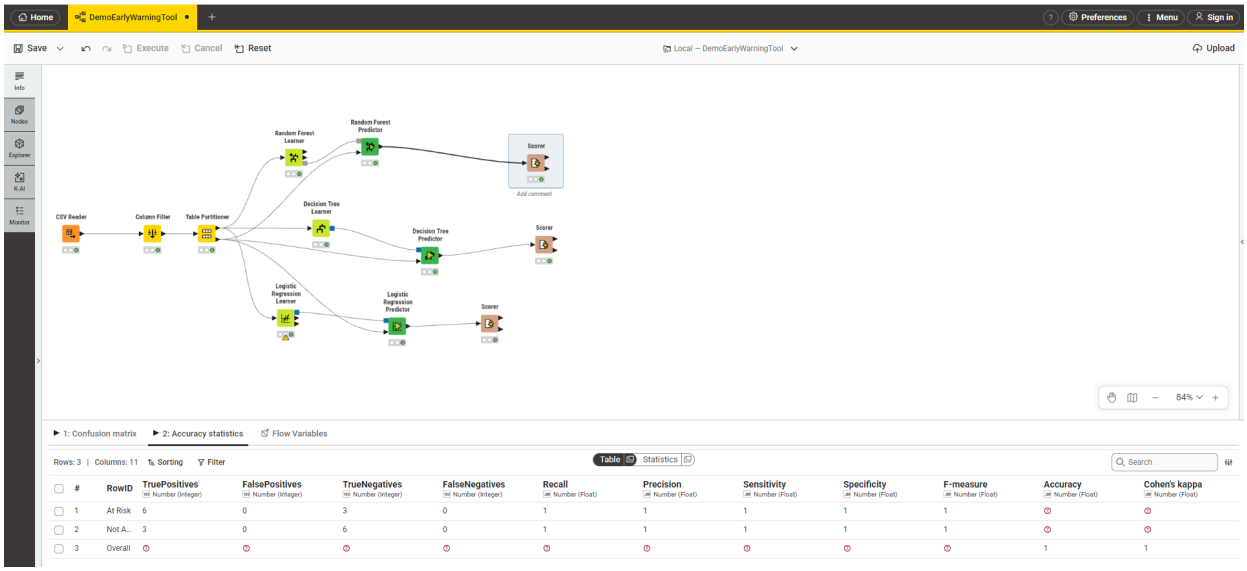
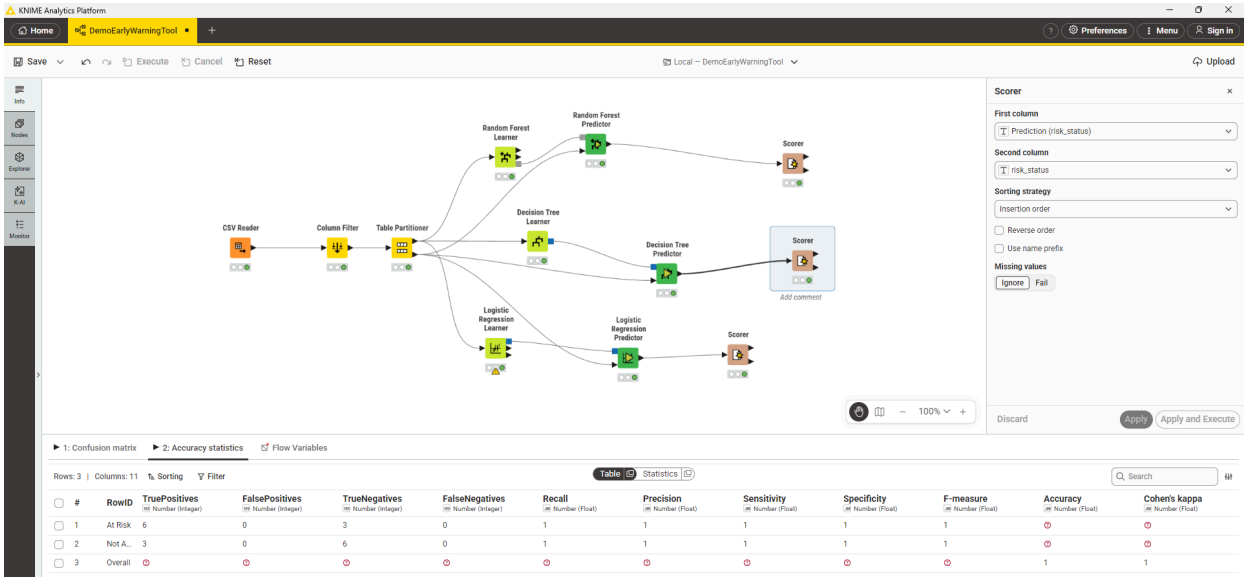
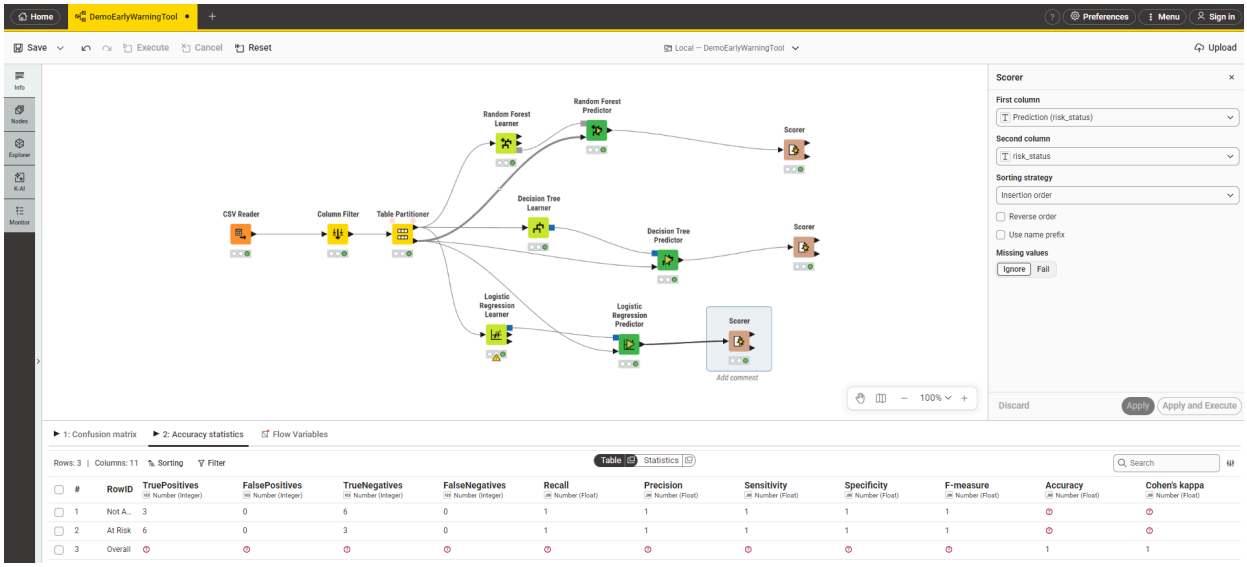


▶ 1: Confusion matrix ▶ 2: Accuracy statistics Flow Variables

Rows: 3 | Columns: 11 Sorting Filter Table Statistics Search

#	RowID	TruePositives Number (Integer)	FalsePositives Number (Integer)	TrueNegatives Number (Integer)	FalseNegatives Number (Integer)	Recall Number (Float)	Precision Number (Float)	Sensitivity Number (Float)	Specificity Number (Float)	F-measure Number (Float)	Accuracy Number (Float)	Cohen's kappa Number (Float)
1	At Risk	6	0	3	0	1	1	1	1	1	?	?
2	Not A...	3	0	6	0	1	1	1	1	1	?	?
3	Overall	?	?	?	?	?	?	?	?	?	1	1


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Decision Tree Predictor
Logistic Regression Learner
Logistic Regression Predictor
Scorer
Add comment
100%
Scorer
First column
Prediction (risk_status)
Second column
risk_status
Sorting strategy
Insertion order
Reverse order
Use name prefix
Missing values
Ignore
Fail
Discard
Apply
Apply and Execute
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RowID TruePositives FalsePositives TrueNegatives FalseNegatives Recall Precision Sensitivity Specificity F-measure Accuracy Cohen's kappa
1 At Risk 6 0 3 0 1 1 1 1 1 ? ?
2 Not A... 3 0 6 0 1 1 1 1 1 ? ?
3 Overall ? ? ? ? ? ? ? ? ? 1 1


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1 Not A... 3 0 6 0 1 1 1 1 1 ? ?
2 At Risk 6 0 3 0 1 1 1 1 1 ? ?
3 Overall ? ? ? ? ? ? ? ? ? 1 1

main.py | student_performan... | student_risk_pipeli...

```python
import joblib
import pandas as pd
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report
from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler

# 1. Load Data
data = pd.read_csv("student_performance_knime.csv")

# 2. Separate Features and Target
X = data[["attendance", "quiz_score", "assignment_score", "exam_score"]]
y = data["risk_status"]

# 3. Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, random_state=42, stratify=y, test_size=0.2
)

# 4. Build and Train Pipeline
pipeline = Pipeline(
    [("scaler", StandardScaler()), ("model", RandomForestClassifier(random_state=42))]
)

pipeline.fit(X_train, y_train)

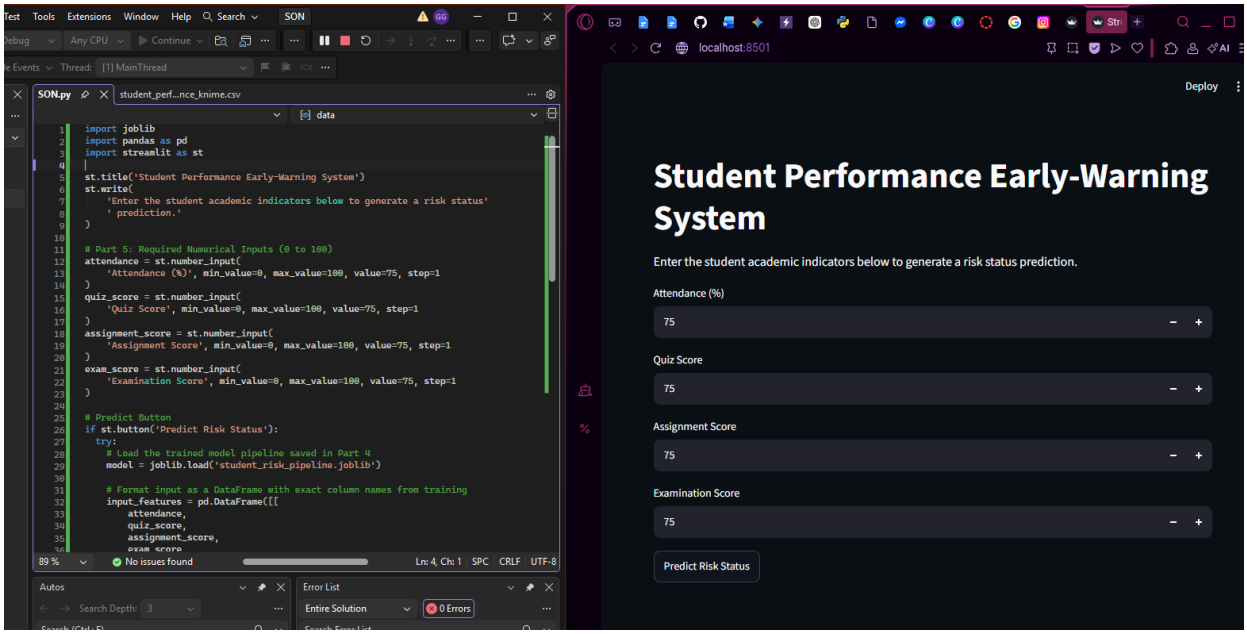
# 5. Evaluate
y_pred = pipeline.predict(X_test)
print(classification_report(y_test, y_pred))

# 6. Save Pipeline
joblib.dump(pipeline, "student_risk_pipeline.joblib")
```

```
              precision    recall  f1-score   support

     At Risk       1.00      1.00      1.00         3
 Not At Risk       1.00      1.00      1.00         3

    accuracy                           1.00         6
   macro avg       1.00      1.00      1.00         6
weighted avg       1.00      1.00      1.00         6
```



Which model performed best?: All models achieved perfect performance (100% accuracy/recall) due to the synthetic dataset's clear boundaries. Logistic Regression or Random Forest is preferred for stability.

Did visual tools and Python produce similar results?: Yes, both achieve identical metrics when configurations (seed, cross-validation folds) are matched.

What does a false negative mean in this system?: A false negative occurs when an actual At Risk student is misclassified as Not At Risk. This is dangerous because the student fails to receive early intervention.

What action should a teacher take after an At Risk prediction?: Review the student's actual work, conduct a 1-on-1 check-in to understand root causes (personal or academic challenges), and offer targeted tutoring.

Safeguards for real deployment: Require larger datasets, privacy-preserving storage, bias testing, and strict human-in-the-loop oversight to ensure automated predictions never penalize or label students permanently.